Questions I would ask stakeholders:

1. Are all the CSV files going to follow the exact same structuring as the example provided, or will I need to handle CSV's that have different structures. For example, entire columns missing or additional/extra columns?
2. What systems (internal? external?) might use our database after the migration?
3. How do we want to approach this migration? No dropped data no matter the reason? Consolidate and simplify the data if possible? Only migrate columns we're interested in?
4. How to handle cases of potential identity fraud? Same health ID, same health provider province. But completely different patient info.
    a. Could just be a clerical error.
    b. Could it be ID reassignment? (Don't know if that actually happens)

For this assignment, I made these **assumptions**/came to these conclusions:

1. Ensure AS MUCH AS possible is moved over, invalid rows gets compiled back into a `.csv` and sent back to the uploader in the same webpage. (if this is supposed to be purely a REST endpoint maybe aggregate the invalid `.csv`'s and send back to the organization that tried to upload them allowing them to find out what needs to be filled in correctly on their end allowing them to try again)
    a. No simplification. But, consolidate the data differently than how it currently is.
2. Load scalability is outside the scope for this assignment (horizontal or vertical).
3. This patient demographic data will be used for most apps (patient portal, EMR, etc.?)
    a. As such frequently fetched data should be in one table to reduce the amount of joins needed. Which should reduce server load and increase speed.
    b. This assumption comes from my recent visits to hospitals and seeing an EPIC system. Older screenshot of an epic system but still makes my point (https://framerusercontent.com/images/w8VHJP3cDXVFLx73G5aMCmBtn4.jpg?width=1677&height=943). Patient address not shown.
4. Patients can not have more than one provincial health plan at once (https://gms.ca/insurance-articles/ultimate-guide-canadian-health-insurance-dual-province-residents)There are no bad actors in this data. If the same Health provider & ID show up with different details it's because the original person is no longer using it and a new person has been assigned the same ID.

With these assumptions I made the following data tables:

Patient: - Patient_ID (PrimKey) - Health_ID - Health_ID_Prov. (abbriv.) - First Name - Middle Name (optional) - Last Name - DOB - Sex - Updated on	Patient_Contact: - Patient_ID - Contact_ID (PrimKey) - Contact_Type - Value	Address: - Patient_ID - Address_ID (PrimKey) - Unit & Street (optional) - City (optional) - Province - Postal (optional)	Uncategorized - Patient_ID - Uncategorized_ID (PrimKey) - Miscellaneous (JSON)

Unhandled tasks due to out of scope:

1. Each uploader would have an identifier allowing us to know which organization they're with.
2. Each organization has their own database using this structure. In case we also need to assign database role permissions that sync up to their role permissions.

What statistics to show?

1. Rows migrated successfully (No issues or warnings) for example 9/10 90%.
2. Notify rows missing critical information. (DOB, both phone & email, first or last name)
    a. Causing the line to be skipped instead
3. how long the import took, in seconds, rounded to two decimals.
4. New, updated and unchanged, what happened to each imported patient as a count.
5. Count of migrated patient's primary provinces

Assumptions were made with links provided above, as well as these:
https://www.314e.com/muspell/health-data-archival-insights/challenges-in-data-migration/
https://guides.rubyonrails.org/active_record_composite_primary_keys.html